Sinan Li

lanceli1416@gmail.com | +44 07301 974443 | www.lance1416.com | linkedin.com/in/sinanli | github.com/lance1416

PROFILE

Oxford **MSc Advanced Computer Science** candidate bridging *quantum computing research*, systems engineering, and full-stack product development. As co-founder and development lead of *IQBank*, architected and shipped a **TypeScript, React, and PostgreSQL** study platform serving **˜100 beta users**, leading cross-functional delivery from MVP through Phase 2. Research focused on *ZX-calculus, fault-equivalence rewrites, and quantum decodability* under Professor Aleks Kissinger; broader interests span formal verification, geometric deep learning, and AI-systems engineering.

EDUCATION

University of Oxford — Oxford, United Kingdom

MSc in Advanced Computer Science — *2025–2026*

- Relevant areas: Quantum Computing, Formal Verification, DBMS, Uncertainty in Deep Learning, Geometric Deep Learning.
- Dissertation: *Investigating the Effect of Fault-Equivalence Rewrites on Quantum Decodability Using ZX-Calculus*.
- Supervisor: Professor Aleks Kissinger.

University of Toronto — Toronto, ON, Canada

Bachelor of Science, Computer Science Specialist; Mathematics Minor — *2021–2025*

- Graduated with **High Distinction**; GPA: **3.94 / 4.0**.
- Relevant areas: Operating Systems, Machine Learning, Artificial Intelligence, Deep Learning.

EXPERIENCE

IQBank — Apr 2025–Present

Co-founder, Development Lead & Product Manager — ***Product + Engineering Leadership***

- Co-founded *IQBank* and led all engineering, launching a study platform that reached **˜100 beta users** with integrated course resources, discussion forums, and instructor-student communication tools.
- Architected and shipped the **full-stack MVP** — TypeScript, React, Express.js, Prisma ORM, and PostgreSQL — with full ownership of backend architecture, REST API design, and data modelling.
- Led an **11-person cross-functional team** through sprint planning, code review, and technical decision-making, coordinating engineering, testing, and UX workstreams.
- Iterated from **MVP to Phase 2** through structured user research, shipping targeted improvements to discussion workflows, accessibility, and content discovery.
- Scoped a *microservices* migration strategy and defined the AI-features roadmap, aligning engineering capacity with product growth targets.

University of Toronto — Jan 2025–Apr 2025

Teaching Assistant, CSC309: Programming on the Web — ***Academic Instruction***

- Designed a **full-stack programming assignment** — spec, grading rubric, and test criteria — requiring students to build a **RESTful** loyalty-program backend with production-grade API and data-modelling constraints.
- Authored end-to-end technical specifications covering **API contract design, server-side logic, and data modelling**, setting clear implementation standards for a project-heavy course.
- Ran weekly office hours guiding students through **REST API design, backend architecture, and debugging** across a wide range of implementation approaches.
- Assessed submissions and exams against detailed rubrics, evaluating correctness, quality, and robustness consistently at scale.
- Conceived the assignment brief later adopted as the foundation of *PointPulse, the CSSU Loyalty Program*.

PROJECTS

KidneyOS — Rust-Based Toy Operating System
2024

University of Toronto; supervised by Professor Kuei Sun

Rust + Systems

- Took sole ownership of the complete **storage subsystem** beneath the VFS layer in a from-scratch **Rust** OS kernel, designing and implementing all components end-to-end.
- Built the **IDE device driver, block layer, and I/O scheduler** from the ground up, bringing full disk access and storage management to the kernel.
- Designed a *block-device abstraction layer* to cleanly decouple hardware-level IDE interactions from the filesystem and upper kernel layers, enabling strict component isolation.
- Solved **async disk I/O, interrupt-driven request scheduling, and direct hardware interfacing** while maintaining UNIX-style modular OS design throughout.
- Coordinated storage integration with the broader kernel team, upholding interface contracts and verifying correctness across subsystem boundaries.

Evogami — Game Design Course Project
2024

University of Toronto

Programming Leadership

- Co-led a **13-person multidisciplinary team** — CS students, music composers, and game designers — through the full development lifecycle of an original game.
- Directed the programming team and personally implemented **core gameplay systems**, including character movement mechanics and multi-stage puzzle interactions.
- Managed programming milestones and **cross-discipline coordination**, bridging technical decisions between designers, audio contributors, and engineers throughout production.
- Established and administered project task-tracking infrastructure, keeping the full team aligned across disciplines and milestones from kickoff to delivery.

AWARDS & HONOURS

Dean's List Scholar | University of Toronto **2021–2025**

- Recognized every academic year for outstanding academic performance.

Later Life Learning Scholarship | University of Toronto **2021, 2024**

- Awarded for outstanding academic performance.

Innis College Exceptional Achievement Award | University of Toronto **2023**

- Awarded for outstanding academic performance.

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, Rust, SQL, C/C++, C#
ML & AI: Deep learning, geometric deep learning, uncertainty in deep learning, model evaluation, PyTorch, NumPy
Research Areas: Quantum computing, quantum error correction, ZX-calculus, diagrammatic reasoning, mathematical computing
Systems & Backend: PostgreSQL, REST APIs, Node.js, Express.js, Prisma ORM, ASP.NET, operating systems
Engineering: Research engineering, technical leadership, experiment design, MVP development, user research, product iteration